

RAMCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Academic Year: 2019-2020(Odd Semester)

INNOVATIVE TEACHING PRACTICES

Degree, Semester& Branch: I Semester B.E. CSE

Course Code & Title: GE8151 Problem Solving and Python Programming

Name of the Faculty member: Mrs.M.Swarna Sudha

Topic: Function definition and its use

Date:13.08.2019

Practice's followed: Think-Pair-Share

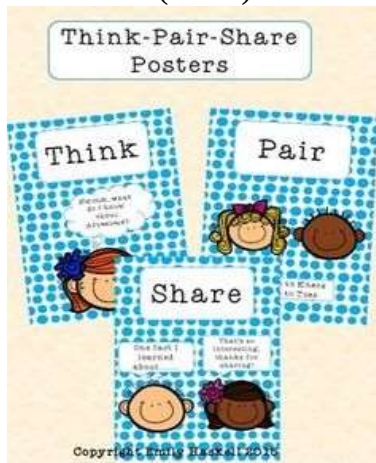
Type of Learning: Active Learning

Learning Objectives:

Objective	Statements
O1	1. To expose the students to develop python programs with functions.

Think-Pair-Share

Collaborative, active learning strategy, in which students work on a problem posed by instructor, first individually (Think), then in pairs or groups (Pair), and finally together with the entire class (Share).



Use of think-pair-share

- It helps the students to think individually about a topic or answer to a question.
- It teaches the students to share ideas with classmates and builds oral communication skills.
- It helps to focus attention and engage students in learning.

Justification for chosen the topic:

The student should clearly understand the concept and usage of function. Functions help to decompose a large program into small segments which makes program easy to understand, maintain and debug. Think – Pair – Share will be an ideal active learning strategy to meet the requirements of the topic.

Implementation of TPS:

This strategy has three steps. First, students think individually about a particular question or scenario. Then they pair up to discuss and compare their ideas. Finally, they are given the chance to share their ideas in a large class discussion.

Think-pair-sharing forces all students to attempt an initial response to the question, which they can then clarify and expand as they collaborate. It also gives them a chance to validate their ideas in a small group before mentioning them to the large group, which may help shy students feel more confident participating

Think:

The instructors posed program where repeated same logic occurs in a program. And also with programs with function which include those codes and execute when needed by calling that function.



Pair up to discuss and compare their ideas

Pair:

The students were asked to identify the difference, also allow them to discuss with their neighbours to know the need of function. The students worked with their neighbours to understand the concept of function and their usage

Share:

The students from each group presented their usage of function and participated in the discussion.



Share their ideas in a large class discussion

Learning Outcomes (LO):

The students were able to know the need of function, and analyse the advantage of usage of function. Apart from charming student actively in learning, this activity help to develop the Individual and team work: And also students were able to improve the communication skill. Finally, the students are able to gain the knowledge and confidence in writing the programs with function.

Objective-PO Mapping

Objective	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	3	3	2						3	2	1	1

Reflective Critique:

Challenges:

- Amount of time was spent to enable the students to understand the concept of think and pair.
- Hesitation in Sharing their ideas in the class

Initiatives to address the problem:

- Students found the activity help them to know the concept of function. Especially when they combine and discuss. Advice the students to know the impact and importance of sharing their views and posting queries related to the topic and made them involve in the activity.
- Assured that activity is not a kind of evaluation methodology to assess.

Post-implementation:

- In a short term of period students had a chance to write many programs with function.
- Coordination level among the students increased
- Students came up with variety of doubts while writing the programs with function.

References:

1. Dr.M.Kaliappan “DTD-DTD for TV schedule” in course “IT6801 Service Oriented Architecture” dated: 10.07.2019. [<https://www.ritrjpm.ac.in/departments/computer-science-engg/innovative-practices.php>]
2. <https://www.teacherspayteachers.com/Product/Think-Pair-Share-2496919>

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